Teacher answer key: Primary 2, Semester 1 (DRAFT)

DRAFT: this answer key is for owner review. Do not give it to pupils. Contact: pypteltd@gmail.com (business email for now, to be changed later)

This key lists the answers and the result of the check on every page. You do not need it for routine marking: a pupil who has a working check has the right answers. To spot-check, compare the check strip with the line "Check result" below.

How the topic tags work

Each entry names the syllabus outcomes the puzzle practises, with the outcome numbers of the Primary 2 content in the Mathematics Syllabus, Primary (2021 edition, pages 33 and 34). The references are for your planning only and are not printed on the pupil pages. "Problem solving" marks a page whose questions come from the kinds found in school tests rather than from one numbered outcome.

Pages in this booklet (Terms 1 and 2)

Strand	Pages
Number and Algebra - Whole Numbers	S1-01, S1-03, S1-04, S1-05, S1-06, S1-07
Number and Algebra - Addition and Subtraction	S1-08, S1-09, S1-10, S1-11
Number and Algebra - Multiplication and Division	S1-12, S1-13, S1-14, S1-15, S1-16
Number and Algebra - Fractions	S1-18, S1-19, S1-20

The page code in the footer, for example S1-07 (P2-17), gives the page in this booklet and, in brackets, its number in the full set of 40.

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S1-01 (P2-01) Hundreds, Tens, Ones 1 star

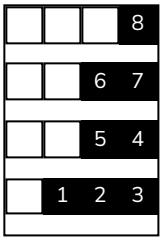
Topic: Place value: hundreds, tens and ones

Syllabus outcomes: Whole Numbers 1.2. Notes on the tag: 1.2 (number notation, representations and place values: hundreds, tens, ones)
Puzzle type: Code-breaker (place value). How the pupil checks: Code word (6 letters).
Answers: 1) 8 hundreds 4 tens 8 ones = 848; 2) 6 hundreds 8 tens 2 ones = 682; 3) 7 hundreds 1 ten 7 ones = 717; 4) 200; 5) 512; 6) 704
Check result: Code word: CLOUDS
Teacher note: Three place-value charts (discs marked 100, 10 and 1), then the value of a digit, a regrouped number (tens that make a hundred or more) and a number with a zero tens place. The wrong numbers a pupil gets by writing the digits in a different order, or by giving the digit instead of its value, have decoy letters in the table or no letter. A box left blank gives no letter and the check cannot see it.

S1-03 (P2-09) Words and Numbers 1 star

Topic: Reading and writing numbers in words and in numerals

Syllabus outcomes: Whole Numbers 1.3. Notes on the tag: 1.3 (reading and writing numbers in numerals and in words)
Puzzle type: Maze / path. How the pupil checks: Route: each step goes to a box whose top words say the numeral at the foot of the box you leave; the line must reach FINISH through the right number of boxes.
Answers: Route (row, column from top-left): (4,2) one hundred and forty-eight / 259 -> (4,3) two hundred and fifty-nine / 538 -> (4,4) five hundred and thirty-eight / 319 -> (3,4) three hundred and nineteen / 454 -> (3,3) four hundred and fifty-four / 746 -> (2,3) seven hundred and forty-six / 883 -> (2,4) eight hundred and eighty-three / 864 -> (1,4) eight hundred and sixty-four / 789
Check result: Route reaches FINISH through 8 boxes; every top word names the number at the foot of the box before it.

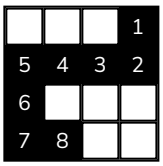


Teacher note: A 4 by 4 grid. Each box has a number in words on top and a numeral at the foot. From a box the pupil moves to a touching box whose top words say the numeral at the foot of the box being left. All numbers are 3-digit. Wrong turns are boxes that look right but are not: digits swapped (341 and three hundred and fourteen), teen and ty mixed up (313 and three hundred and thirty), a zero in the wrong place (305 and three hundred and fifty). 2 wrong turns can be entered from the route and lead nowhere, so the box count and the words matter. Pupils read numbers in words and match them with numerals; writing number words by hand is not checked by this page.

S1-04 (P2-12) Odd or Even 1 star

Topic: Odd and even numbers: even numbers

Syllabus outcomes: Whole Numbers 1.6. Notes on the tag: 1.6 (odd and even numbers)
Puzzle type: Maze / path. How the pupil checks: Route: the line must reach FINISH through the right number of boxes, every box even.
Answers: Route (row, column from top-left): (1,4) 454 -> (2,4) 748 -> (2,3) 336 -> (2,2) 308 -> (2,1) 418 -> (3,1) 388 -> (4,1) 396 -> (4,2) 774
Check result: Route reaches FINISH through 8 boxes; every box on it is even.



Teacher note: A 4 by 4 grid of sixteen 3-digit numbers. The route follows even numbers: the pupil looks at the last digit. The other boxes hold the other kind of number; 2 extra even boxes are dead ends. Counting the boxes shows a line that takes a shortcut.

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S1-05 (P2-10) Digit Cards 2 stars

Topic: Numbers to 1000: make, compare and order numbers from digit cards

Syllabus outcomes: Whole Numbers 1.2, Whole Numbers 1.4, Whole Numbers 1.6. Notes on the tag: 1.2, 1.4 (place values; comparing and ordering numbers); odd and even numbers 1.6
Puzzle type: Code-breaker (digit cards). How the pupil checks: Code word (6 letters).
Answers: 1) smallest odd: 285; 2) second smallest: 275; 3) second greatest: 401; 4) greatest: 621; 5) greatest even: 718; 6) smallest: 509
Check result: Code word: CURFEW
Teacher note: Six sets of three digit cards: greatest, smallest, second greatest, second smallest, greatest even and smallest odd. A zero card cannot be the hundreds digit. The decoy letters stand for the other arrangements of the same cards, so a pupil who makes a different number from the same cards still gets a letter, but not the right one. A box left blank gives no letter and the check cannot see it.

S1-06 (P2-11) Number Patterns 2 stars

Topic: Number patterns: counting on and back in 2s, 3s, 4s, 5s, 10s and 100s

Syllabus outcomes: Whole Numbers 1.1, Whole Numbers 1.5. Notes on the tag: 1.1, 1.5 (counting in tens and hundreds; patterns in number sequences)
Puzzle type: Code-breaker (sequences). How the pupil checks: Code word (6 letters).
Answers: 1) +10: missing 580; 2) +100: missing 790; 3) +3: missing 194; 4) -100: missing 630; 5) -5: missing 704; 6) +4: missing 326
Check result: Code word: FATHOM
Teacher note: Six rows, each counting on or back by 2, 3, 4, 5, 10 or 100 (one rule per row, all different). Several rows cross a hundred, so the hundreds digit changes. The decoy letters stand for the usual slips: the step added twice, or only one digit changed. A box left blank gives no letter and the check cannot see it.

S1-07 (P2-17) Mystery Numbers 2 stars

Topic: 3-digit numbers: digits, odd and even, more and less

Syllabus outcomes: Whole Numbers 1.2, Whole Numbers 1.4, Whole Numbers 1.6. Notes on the tag: 1.2, 1.4, 1.6 (place values; comparing and ordering numbers; odd and even numbers)
Puzzle type: Number riddles. How the pupil checks: Code word (5 letters).
Answers: 1) 498; 2) 988; 3) 202; 4) 918; 5) 773
Check result: Code word: VALET
Teacher note: Every riddle has three or four clues and every clue is needed: with one clue left out, more than one number fits. A pupil who stops after the first clues gets a number that fits some of them; testing every clue removes the others. Clues used: more than, less than, a digit of a place, odd, even, one digit more than another, two digits the same. The decoy letters stand for numbers that fit most of the clues. A box left blank gives no letter and the check cannot see it.

S1-08 (P2-02) Column Sums 1 star

Topic: Add and subtract in columns, 3-digit numbers

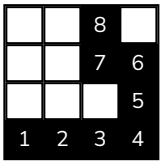
Syllabus outcomes: Addition and Subtraction 2.1. Notes on the tag: 2.1 (addition and subtraction algorithms, up to 3 digits)
Puzzle type: Code-breaker (written methods). How the pupil checks: Code word (6 letters).
Answers: 1) 304 + 219 = 523; 2) 433 + 124 = 557; 3) 142 + 294 = 436; 4) 722 - 216 = 506; 5) 746 - 594 = 152; 6) 648 - 124 = 524
Check result: Code word: DAINTY
Teacher note: Six vertical sums within 1000: three additions (no renaming, ones renamed, tens renamed) and three subtractions (no renaming, ones renamed, tens renamed). The decoy letters stand for the answers a pupil gets by not renaming (adding or subtracting digit by digit). A box left blank gives no letter and the check cannot see it.

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S1-09 (P2-03) Mental Jumps 1 star

Topic: Mental sums: 3-digit numbers with ones, tens and hundreds

Syllabus outcomes: Addition and Subtraction 2.2. Notes on the tag: 2.2 (mental calculation: a 3-digit number with ones, tens or hundreds)
Puzzle type: Maze / path. How the pupil checks: Route: the line must reach FINISH through the right number of boxes, every box making the target.
Answers: Route (row, column from top-left): (4,1) 455 + 200 -> (4,2) 653 + 2 -> (4,3) 855 – 200 -> (4,4) 660 – 5 -> (3,4) 255 + 400 -> (2,4) 625 + 30 -> (2,3) 647 + 8 -> (1,3) 585 + 70
Check result: Route reaches FINISH through 8 boxes; every box on it makes 655.



Teacher note: A 4 by 4 grid of sixteen mental sums with a 3-digit number and ones, tens or hundreds. The boxes off the route make answers near 655 (635, 654, 656, 755); a slip in adding or taking away sends the pupil there. 2 extra boxes that do make 655 are dead ends. Counting the boxes shows a line that takes a shortcut.

S1-10 (P2-13) Missing Digits 2 stars

Topic: Add and subtract in columns: find the missing digit

Syllabus outcomes: Addition and Subtraction 2.1. Notes on the tag: 2.1 (addition and subtraction algorithms, up to 3 digits)
Puzzle type: Code-breaker (written methods). How the pupil checks: Code word (6 letters).
Answers: 1) 532 - 222 = 310: digit 0; 2) 604 + 382 = 986: digit 3; 3) 892 - 143 = 749: digit 4; 4) 153 + 114 = 267: digit 1; 5) 123 + 459 = 582: digit 5; 6) 493 - 204 = 289: digit 2
Check result: Code word: CREAMY
Teacher note: Six column sums with one digit missing (three additions, three subtractions, the missing digit in the top number, the bottom number or the answer). Each has exactly one digit that makes it true. The pupil writes the missing digit, then looks it up. A digit that is wrong either has a decoy letter or no letter. A box left blank gives no letter and the check cannot see it.

S1-11 (P2-18) Story Sums to 1000 2 stars

Topic: Two-step stories with numbers to 1000

Syllabus outcomes: Addition and Subtraction 2.1, Addition and Subtraction 2.2 (problem solving). Notes on the tag: 2.1, 2.2 (addition and subtraction algorithms up to 3 digits; mental calculation)
Puzzle type: Code-breaker (two-step stories). How the pupil checks: Code word (6 letters).
Answers: 1) 404 then 180; 2) 629 then 427; 3) 624 then 755; 4) 559 then 521; 5) 365 then 520; 6) 316 then 569
Check result: Code word: SPONGE
Teacher note: Six stories, each with two steps within 1000: add then take away, take away twice, add twice, a 'more than' story and a 'fewer than' story (the child who has fewer is the one named first; in all means both children), and a story in which one number changes before the two are compared. The decoy letters stand for the answer of the first step only. A box left blank gives no letter and the check cannot see it.

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S1-12 (P2-04) Equal Groups 1 star

Topic: Multiplication: tables of 2, 3, 4, 5 and 10

Syllabus outcomes: Multiplication and Division 3.1, Multiplication and Division 3.4. Notes on the tag: 3.1, 3.4 (multiplication tables of 2, 3, 4, 5 and 10; multiplying within the tables)
Puzzle type: Code-breaker (equal groups). How the pupil checks: Code word (6 letters).
Answers: 1) 7; 2) 20; 3) 9; 4) 10; 5) 18; 6) 16
Check result: Code word: LENGTH
Teacher note: Two arrays of objects (count rows and columns or count all), a bare multiplication, a groups-of story, a missing factor and a table-top story. Every product uses a table of 2, 3, 4, 5 or 10 and stays within 50. The decoy letters stand for adding the two numbers instead of multiplying. A box left blank gives no letter and the check cannot see it.

S1-13 (P2-05) Share It Out 1 star

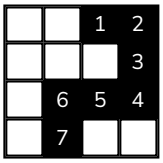
Topic: Division: sharing and grouping, the sign ÷

Syllabus outcomes: Multiplication and Division 3.2, Multiplication and Division 3.4. Notes on the tag: 3.2, 3.4 (use of ÷; dividing within the multiplication tables)
Puzzle type: Code-breaker (division). How the pupil checks: Code word (6 letters).
Answers: 1) 6 ÷ 2 = 3; 2) 32 ÷ 4 = 8; 3) 12 ÷ 3 = 4; 4) 12 ÷ 2 = 6; 5) 70 ÷ 10 = 7; 6) 27 ÷ 3 = 9
Check result: Code word: NUMBER
Teacher note: Division by 2, 3, 4, 5 and 10 within the tables: two bare sums with the sign ÷, equal sharing, grouping into boxes, a 'how many 3s in 12' item and one story where the sharer also gets a share ('shares with 3 friends' means 4 people). The decoy letters stand for subtracting or for using the number of friends instead of the number of people. A box left blank gives no letter and the check cannot see it.

S1-14 (P2-14) Times Table Maze 1 star

Topic: Multiplication and division facts: tables of 2, 3, 4, 5 and 10

Syllabus outcomes: Multiplication and Division 3.1, Multiplication and Division 3.2, Multiplication and Division 3.5. Notes on the tag: 3.1, 3.5 (tables of 2, 3, 4, 5, 10; mental calculation within the tables); 3.2 ÷
Puzzle type: Maze / path. How the pupil checks: Route: the line must reach FINISH through the right number of boxes, every box making 10.
Answers: Route (row, column from top-left): (1,3) 1 × 10 -> (1,4) 20 ÷ 2 -> (2,4) 2 × 5 -> (3,4) 40 ÷ 4 -> (3,3) 100 ÷ 10 -> (3,2) 10 × 1 -> (4,2) 50 ÷ 5
Check result: Route reaches FINISH through 7 boxes; every box on it makes 10.



Teacher note: A 4 by 4 grid of sixteen multiplication and division facts from the tables of 2, 3, 4, 5 and 10. The route follows the boxes that make 10. Wrong boxes have answers close to 10. 2 extra boxes that do make 10 are dead ends. Counting the boxes shows a line that takes a shortcut.

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S1-15 (P2-19) Fact Families 2 stars

Topic: Multiplication and division: finding the missing number, the loop that closes

Syllabus outcomes: Multiplication and Division 3.2, Multiplication and Division 3.3, Multiplication and Division 3.4. Notes on the tag: 3.2, 3.3 (use of ÷; relationship between multiplication and division); 3.4
Puzzle type: Chain loop (multiplication tiles). How the pupil checks: Loop: the number in an empty circle must match the circle it is joined to by =, all the way round and back to tile 1.
Answers: Empty circles: tile 1 right = 6, tile 2 left = 6, tile 3 right = 3, tile 4 right = 2, tile 5 right = 7, tile 6 right = 2. Tiles (top = left × right): 12 = 2 × 6, 24 = 6 × 4, 12 = 4 × 3, 6 = 3 × 2, 14 = 2 × 7, 14 = 7 × 2
Check result: Every touching pair matches, all the way round.
Teacher note: Six tiles, each a product at the top and its two factors below. One factor of every tile is empty; the pupil finds it by dividing the top by the other factor (the relationship between multiplication and division) and may check by multiplying back. The right factor of a tile is the same number as the left factor of the next tile, and the right factor of the last tile is the first tile's left factor. So every number the pupil writes is tested against a matching circle: a wrong number breaks the loop. At most one link has an empty circle on both sides. In every tile at least one factor is from the tables of 2, 3, 4, 5 and 10; all products are within 60.

S1-16 (P2-20) True or False Tables 2 stars

Topic: Multiplication and division facts: true or false

Syllabus outcomes: Multiplication and Division 3.1, Multiplication and Division 3.3, Multiplication and Division 3.4, Multiplication and Division 3.5. Notes on the tag: 3.1, 3.4, 3.5 (tables of 2, 3, 4, 5 and 10; multiplying and dividing within the tables; mental calculation)
Puzzle type: True or false table. How the pupil checks: Word from the letters of the right choices (6 letters).
Answers: 1) T (4 × 5 = 20); 2) T (10 groups of 8 is 80.); 3) F (36 ÷ 4 = 11); 4) F (If 3 × 9 = 27, then 27 ÷ 9 = 2.); 5) T (5 × 6 = 30); 6) F (17 is in the 4 times table.)
Check result: Word: WALNUT
Teacher note: Six statements from the tables of 2, 3, 4, 5 and 10, including one about a times table and one that links a multiplication to a division. Each row has a T choice and an F choice, each with a letter. The letters of the right choices, read down the rows, spell the word. Every other way of choosing was tested against a very large word list: none spells a word. A box left blank gives no letter and the check cannot see it.

S1-18 (P2-15) Parts of a Whole 1 star

Topic: Fractions: parts of a whole, shaded parts

Syllabus outcomes: Fractions 1.1, Fractions 1.2. Notes on the tag: 1.1, 1.2 (fraction as part of a whole; notation and representations of fractions)
Puzzle type: Code-breaker (fractions). How the pupil checks: Code word (6 letters).
Answers: 1) 1/12; 2) 2/3; 3) 7/12; 4) 1/4; 5) 4/5; 6) 1/5
Check result: Code word: FROZEN
Teacher note: Six figures (bars, circles and grids) cut into equal parts, with some parts shaded black. The pupil writes the shaded part as a fraction (top box: shaded parts, bottom box: all the equal parts). No fraction is one that can be written in simpler parts (no 2/4 or 3/6): equivalent fractions are Primary 3. The decoy letters stand for reading the unshaded parts, or counting one part too many. A box left blank gives no letter and the check cannot see it.

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S1-19 (P2-21) Fraction Order 2 stars

Topic: Fractions: comparing and ordering like fractions and unit fractions

Syllabus outcomes: Fractions 1.2, Fractions 1.3. Notes on the tag: 1.2, 1.3 (notation of fractions; comparing and ordering unit fractions and like fractions)
Puzzle type: Ordering (lettered cards). How the pupil checks: Letters of the cards in the order asked spell a word (6 letters).
Answers: Set A (smallest first): B = 3/9, A = 5/9, N = 8/9; Set B (greatest first): D = 1/2, I = 1/5, T = 1/6
Check result: Word: BANDIT
Teacher note: Set A has three like fractions (same denominator): the larger the top number, the greater the fraction. Set B has three unit fractions (tops all 1): the larger the bottom number, the smaller the fraction, so greatest first means the smallest bottom number first. The letters, read in the order asked, spell a word with exactly one arrangement that is a word, so no wrong order of the six cards spells a word (every wrong order was tested against a very large word list). A box left blank gives no letter and the check cannot see it.

S1-20 (P2-22) Fraction Sums 2 stars

Topic: Fractions: adding and subtracting like fractions within one whole

Syllabus outcomes: Fractions 2.1. Notes on the tag: 2.1 (adding and subtracting like fractions within one whole)
Puzzle type: Code-breaker (fractions). How the pupil checks: Code word (6 letters).
Answers: 1) 1/6; 2) 4/5; 3) 11/12; 4) 9/12; 5) 3/8; 6) 7/12
Check result: Code word: ANSWER
Teacher note: Six items with like fractions (denominators up to 12): an addition, two subtractions, 'what makes 1 whole', one whole minus a fraction and a story. The bottom number is printed; the pupil writes the top number. The decoy letters stand for the wrong operation, or for the part left over. A box left blank gives no letter and the check cannot see it.

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